SALMETEROL + FLUTICASONE

Never LABA alone in asthma

Maintenance only

CHEMICAL NAME Salmeterol xinafoate: (RS)-2-(hydroxymethyl)-4-{1-hydroxy-2-[6-(4-phenylbutoxy)hexylamino]ethyl}phenol + fluticasone propionate

CLASS Combination inhaler: long-acting β2 agonist (LABA) + inhaled corticosteroid

TRADE NAMES Seretide · Advair · Sirdupla · AirFluSal

THERAPEUTIC USES / INDICATIONS

- Asthma maintenance when not controlled on ICS alone.
- COPD with frequent exacerbations (500/50 Accuhaler).

PHARMACODYNAMICS

- Salmeterol: **β2 agonist** with long lipophilic tail → 12 h bronchodilation.
- Fluticasone: anti-inflammatory; steroid and LABA act synergistically (↑ β2-receptor expression).

ADMINISTRATION

- **Accuhaler (dry powder):** open, slide lever, inhale fast and deep; twice daily.
- **Evohaler (pMDI):** with spacer; shake before use.
- Rinse mouth after each use.
- Not for acute relief (salmeterol onset 15–30 min).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Oral thrush, hoarseness.
- Headache, tremor, palpitations.
- Muscle cramps.

ADVERSE EFFECTS (SERIOUS)

- Pneumonia (COPD).
- Paradoxical bronchospasm.
- Hypokalaemia, arrhythmias (high doses).
- Adrenal suppression (high doses).

DOSAGES

- **Asthma (Accuhaler):** 1 inhalation 100/50, 250/50 or 500/50 mcg twice daily (fluticasone/salmeterol).
- **Evohaler:** 2 puffs 50/25, 125/25 or 250/25 twice daily.
- **COPD:** 500/50 mcg twice daily.
- **Children 4–12 yr:** 100/50 twice daily.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity (incl. milk protein — lactose in Accuhaler).
- **Caution:** cardiovascular disease, QT prolongation, TB, diabetes.

KEY INTERACTIONS

- **Ritonavir, ketoconazole:** ↑ both drugs (Cushing's, QT).
- **β-blockers:** antagonism.
- **Other LABAs:** avoid duplication.

PHARMACOKINETICS

- **Salmeterol onset:** 15–30 min; duration 12 h.
- Both metabolised by CYP3A4.
- Low systemic bioavailability.

NURSING MANAGEMENT

- Check that patient also has a SABA reliever.
- Assess technique, symptom control, exacerbations.
- Inspect mouth; monitor for pneumonia in COPD.
- Step down when stable.

PATIENT EDUCATION

- Use twice daily every day; this does not relieve sudden attacks; use your blue reliever.
- Rinse mouth after use.
- Check the dose counter.
- Report increasing reliever use.

KEY POINTS

- LABA + ICS twice daily.
- LABA never without ICS in asthma.
- Not a reliever.
- Rinse mouth; pneumonia risk in COPD.